

MOHSIN ALI NISAR

Senior Software Engineer & Solutions Architect

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PROFESSIONAL SUMMARY

Senior Software Engineer and Solutions Architect with over 15 years of experience engineering mission-critical enterprise systems across supply chain logistics, **AI Video Surveillance**, **Medical Billings**, **USAID** monitoring and **Fleet Management** real-time tracking platforms. Established as a technical anchor and engineering lead who solves high-impact production anomalies that stop other teams, while actively mentoring cross-functional groups to drive deep technical execution.

CORE SKILLS

Languages: **C# (.NET Core 8+)**, **Python**, **JavaScript/TypeScript**, **C/C++**

Architecture: **Microservices**, **Event-Driven Architecture**, **DDD**, **CQRS**, **Clean/Onion Architecture**, **gRPC**

AI/ML: **YOLOv8**, **OpenVINO**, **LangChain (RAG)**, LLM Orchestration (**Ollama/OpenAI**), **ML.NET**, **Azure Cognitive Services**

Cloud & Data: **Azure** (Functions, **Service Bus**, **AKS**, Blob Storage), **SQL Server**, **PostgreSQL**, **NoSQL**, **Elasticsearch**

Real-Time & Media: **WebRTC**, **HLS/WHEP**, **SignalR**, **MQTT**, **FFmpeg**

Web: **ASP.NET MVC Core**, **Blazor**, **Angular**, **React**, **OAuth 2.0/OpenID Connect**, **JWT**

DevOps: **CI/CD (GitHub Actions, Azure DevOps)**, **Docker**, **Kubernetes**, **Agile/Scrum**

PROFESSIONAL EXPERIENCE

Senior Software Engineer — SMASH / BlueEye (Remote, USA)

Jul 2024 – Present

- **Architected Legacy Monolith Migration:** Led strategic decoupling of a legacy **Node.js** platform into a **.NET Core** microservices ecosystem on **Azure** (bounded contexts: video ingestion, AI analytics, notifications) leveraged **.NET** multi-threading to cut production runtime errors.
- **Low-Latency Streaming Pipeline:** Directed architecture of a **WebRTC/HLS/WHEP** livestreaming engine using **Azure Service Bus** and Event Grid, successfully isolated camera-telemetry spikes from core workflows and cleared engineering team delivery deployment bottlenecks.
- **AI Anomaly Detection:** Engineered command-and-control layers parsing **VAPIX** events via **gRPC**, implemented **Chain-of-Thought** reasoning logic to optimize alert reliability and eliminate noisy false positives for operators.
- **Edge Performance Engineering:** Converted heavy **YOLO** models to **OpenVINO** IR format and optimized **Python** processing pipelines integrated with **.NET** to achieve smooth, production-grade real-time frame streaming on hardware-constrained edge devices.
- **Secure IoT Audio:** Developed an **MQTT**-driven 'talk-down' audio sub-system utilizing the **Orchid API**, **FFmpeg** processing, and **JWT** security keys to scale active responses over distributed camera footprints.

Senior Software Engineer — GlobalTranz (Remote, USA)

Mar 2021 – Jul 2024

Started on the platform stabilization team, then transitioned to lead the microservices modernization of the platform.

- **Legacy Recovery & Stabilization:** Took full ownership of an undocumented, multi-vendor 3rd-party logistics suite, reverse-engineered code to produce foundational architecture docs, stabilized Accounting/Auditing modules, and neutralized severe inline SQL injection vulnerabilities.
- **Fin-Tech & Event-Driven Ledger:** Governed the transactional Carrier Rate & Accounting Center, engineered a serverless Quick Pay workflow processing immutable events (Tender Accepted, Invoice Audited) via **Azure Service Bus** to compute dynamic surcharges and complete automated ACH bank settlements within 24 hours.
- **Automated Audit Architecture:** Built domain-oriented **C#/.NET Core** microservices (**Clean Architecture**) deployed via **Docker** on **Kubernetes (AKS)**, automated freight discrepancy routing by leveraging **Azure Service Bus Dead-Letter Queues** for hands-free invoice dispute generation.
- **Production Bug Rescue:** Isolated and permanently fixed a legacy, production-only UVB duplication bug that had baffled prior engineering efforts by dry-ran raw production log state machines to diagnose an insert-vs-update logic flaw, protecting critical database integrity.

- **Data Layer Optimization:** Diagnosed silent **Elasticsearch** indexing/hydration failures and built robust UI error handling fallbacks, while refactoring the platform's largest legacy stored procedures to slash their logic footprints in half while multiplying query execution performance.
- **Diagnostic & Observability Engineering:** Remediated a highly degraded accounting module by reconstructing query execution plans, purging invalid joins, and mapping correct data types, while implementing **Correlation IDs** via **Azure Application Insights** for distributed tracing and served as the tier-3 escalation anchor for critical tickets.



Senior Software Engineer — Cube Health Care System (Islamabad, Hybrid)

Aug 2020 – Jan 2021

- **Healthcare Compliance Modernization:** Upgraded core Revenue Cycle Management (RCM) billing portals using **.NET Core** and **Angular**, designed Audit Logging, Exception Tracking, and Session Monitoring blocks to strictly comply with healthcare data privacy laws.
- **Real-Time Micro-Services:** Automated an instant notification framework for inbound user queries to remove support latency built modular portal extensions that safely decoupled legacy code dependencies.



Sr. Software Engineer / System Architect — Management Systems International

Jul 2014 – Nov 2019

- **Enterprise Cloud Platforms:** Engineered a national USAID project tracking ecosystem (IMEC) deploying secure **RESTful APIs**, complex **Enterprise Search** indexes, and automated data collection pipelines.
- **GIS & Emergency Dashboards:** Supervised the construction of multi-tier situational awareness centers while launched the Pakistan Interactive Maps II GIS platform with **D3.js**, **ArcGIS Server**, and **Leaflet.js** for country-scale visual monitoring.

EARLY CAREER HISTORY

 Sr. Software Engineer, Team Lead & Architect — BrightSpyre.com (2013–2014): Migrated field-data workflows to ASP.NET MVC/Google Maps built **Python/Django** remote team payroll tracking applications.

 Full Stack Software Engineer & Architect — Xpert Technologies (2012–2013): Ported desktop telematics software into an **Onion Architecture** web system featuring real-time vehicle geo-fencing frameworks.

 Senior Software Engineer, Technical Lead — Advance I-Tech Solutions (2011–2012): Ran full SDLC design for a modular enterprise ERP platform spanning unified HRM, Payroll, and Security workflows.

EDUCATION

 Bachelor of Computer Science — Hazara University, 2009